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Implementation for Fetal ECG Detection from Multi-channel Abdominal Recordings with 2D Convolutional Neural Network

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Abstract

A convolutional neural network (CNN)-based approach for fetal ECG detection from the abdominal ECG recording is proposed. The flow contains a pre-processing phase and a classification phase. In the pre-processing phase, short-time Fourier transform is applied to obtain the spectrogram, which is sent to 2D CNN for classification. The classified results from multiple channels are then fused and high detection accuracy up to 95.2% is achieved and the CNN-based approach outperforms the conventional algorithm. The hardware of this fetal ECG detector composed of the spectrogram processor and 2D CNN classifier is then implemented on the FPGA platform. Because the two dimensions of the spectrogram and the kernel are asymmetric, a pre-fetch mechanism is designed to eliminate the long latency resulted from data buffering for large-size convolution. From the implementation results, it takes 20258 clock cycles for inference and almost 50% computation cycles are reduced. The power consumption is 12.33mW at 324KHz and 1V for real-time operations. The implementation demonstrates the feasibility of real-time applications in wearable devices.

Keywords Fetal electrocardiogram · Convolutional neural network · Short-time Fourier transform

1 Introduction

Fetal arrhythmia and fetal atrioventricular (AV) block are two common reasons for fetal demise [1, 2]. Abdominal electrocardiogram recording is essential to diagnose congenital heart defects and to assess health condition of the fetus during pregnancy. The non-invasive approach is often preferred in order to prevent from infection and injury. The fetal electrocardiogram (fECG) can be acquired by the surface electrodes placed on the mother's abdomen. However, the measured signal is a mixture of various components, such as the maternal electrocardiogram (mECG), disturbance from maternal muscle and respiration activity as well as fetal movement, and noise. The distribution of most of the undesired disturbances in spectrum is usually quite different from fECG and thus can be removed by frequency-domain processing. However, the mECG not only occupies a similar frequency band but also has larger magnitude [4] and thus removal of this major interfering component becomes a challenging task. During these years, various signal processing techniques are developed to extract or detect the fECG. Furthermore, with the prosperous development of wearable devices, continuous monitoring of the physiological signals is feasible. If the fECG extraction or detection techniques can be realized in wearable devices, it will benefit clinical diagnosis at the early stage.

Blind source separation (BSS) has been employed for fECG extraction, which achieves better performance than matched filtering and correlation [3]. Independent component analysis (ICA), one of the famous algorithms for BSS, has been widely adopted to decompose the mixed signal. In [5], fast ICA has been shown to outperform Infomax ICA in terms of computation time and complexity. In [7], besides fast ICA, undecimated wavelet transform and fast Fourier transform as well as inverse Fourier transform are employed in post-processing to denoise so as to enhance the detection accuracy. Fast ICA with maternal R-peak suppression

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algorithm has been proposed in [6]. The authors apply ICA on compressed signals and utilize ℓ_p-regularized recursive least squares algorithm for reconstruction in [8]. However, the ordering of extracted components varies for each trial with the ICA algorithm [5], and thus further recognition is required. On the other hand, time-frequency analysis also plays an important role [9, 10] in ECG related research. Smoothed pseudo Wigner–Ville distribution (SPWVD) is used in [9] while short time Fourier transform (STFT) is applied in [10] for fetal ECG extraction.

Due to the development of machine learning and artificial intelligence, there are some works which formulate the related issues as classification problems. The electrocardiogram is partitioned into waveform segments and the event in each segment is detected. The advanced classifier, one-dimensional convolutional neural network (CNN), is adopted to time-domain ECG in [11] to detect the events of ventricular ectopic beats and supraventricular ectopic beats. Furthermore, two-dimensional CNN is applied to time-domain ECG waveform segment in [12] for heart beat classification. Similarly STFT is considered in [13] and [14] to exploit more information. Then, two-dimensional CNN dealing with STFT spectrogram can be seen in [15] for ECG arrhythmia classification. Following the success of ECG classification, the authors in [16] reformulate the fetal ECG recognition problem. The segments of abdominal ECG waveform are classified into four classes, including F-event for fetal QRS (fQRS) complex, M-event for maternal QRS (mQRS) complex, MF-event for co-existence of fQRS and mQRS, and N-event for non-existence of any QRS complex. Support vector machine (SVM), K-nearest neighbor (KNN), and Bayesian network (BN) are used for classification. In [18], we first applied two-dimensional CNN to abdominal ECG spectrogram for fetal ECG detection and the detection accuracy has been enhanced. The detection of the fetal ECG event could help the analysis and diagnosis of fetal health condition, as discussed in [17].

In this paper, we modify the kernel of the two-dimensional CNN in [18] and develop an asymmetrical kernel to deal with the spectrogram of abdominal ECG after STFT. The asymmetric kernel greatly reduces the complexity of CNN and achieves the same performance as the symmetric one. The whole fECG detector is then implemented on the FPGA to demonstrate its feasibility for real-time processing and to show the applicability in low-power wearable devices.

In the following, the signal processing flow and CNN structure are illustrated in Section 2. The detection performance is discussed in Section 3. Design of hardware architecture is delineated in Section 4. Section 5 provides the implementation results and Section 6 concludes the paper.

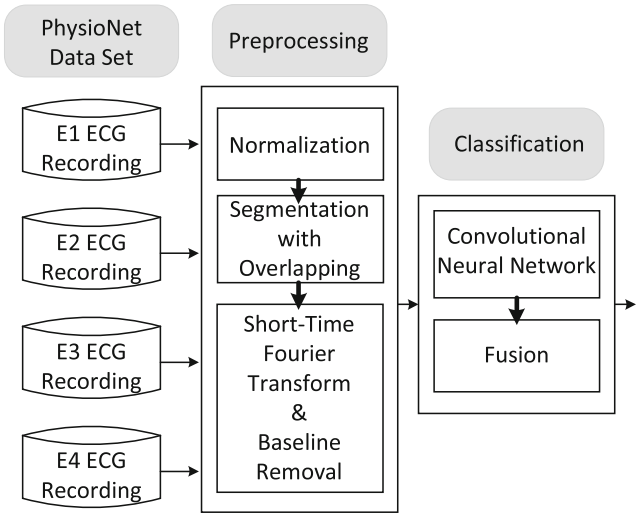


Figure 1 Signal processing flow.

2 Signal Processing Flow

The signal processing flow of fetal ECG detection is shown in Figure 1. The multi-channel abdominal ECG recordings in PhysioNet [19] are used, which contain four sets of signals from maternal abdomen and one reference signal in 5 minutes. The analog-to-digital converter (ADC) is 16 bits wide with 1KHz sampling frequency. Then, preprocessing is applied before classification. The detailed operations are addressed in the following.

2.1 Preprocessing

In the pre-processing stage, the signal amplitude is fist normalized to $[-1, 1]$. Then, the recording is partitioned into segments each having 270ms long with two 10ms overlapping regions at two ends to prevent from the fragmented QRS complex. Applying short-time Fourier transform (STFT), we have the time-frequency representation $\mathbf{X}_i$, and its (n, k)th element is given by

$$\mathbf{X}_i(n,k) = \sum_{m=0}^{M-1} x_i(m + n\Delta)w(m)e^{-j2\pi km/M} \tag{1}$$

where $x_i(m)$ is the abdominal signal of lead i, $\Delta(= 18)$ is the hop size, $w(m)$ is the Hamming window described by

$$w(m) = \begin{cases} 0.538 - 0.462\cos(\frac{2\pi m}{\tilde{M}-1}) & 0 \le m \le \tilde{M}-1 \\ 0 & else \end{cases}. \tag{2}$$

The window size $\tilde{M}$ is set to 36 and the FFT size M is 256. Consequently, the dimension of the spectrogram is 128×14.

Figure 2 shows the waveform segments corresponding to the four events, namely F event for fQRS complex, M event for mQRS complex, MF-event for co-existence of fQRS and mQRS, and N-event for non-existence of any QRS complex. Each segment is 270ms long. Figure 3 provides the spectrogram after STFT associated with the four events in Figure 2. We can see that the QRS complex in the time domain generates frequency components around 10-160Hz. To get rid of the influence from baseline wandering, the DC component is removed.

2.2 Classification

Since good classification results are obtained by CNN in [11] as well as [15] for ECG arrhythmia detection, we then employ 2D CNN for the event classification so that the fetal ECG can be recognized. Figure 4 depicts the entire structure of the 2D CNN with two convolutional layers, two pooling layers and one fully-connected layer.

Without loss of generality, we drop the subscript i for lead i here. Let $\mathbf{Z}_q^{(0)} = \mathbf{X}$. In the first convolutional layer, six neurons are used. Generally, the feature map of the pth neuron output can be described

$$\mathbf{Y}_p^{(l)} = b_p^{(l)} + \sum_{q=0}^{P_{l-1}-1} Conv_{2D}\left(\mathbf{W}_{p,q}^{(l-1)}, \mathbf{Z}_q^{(l-1)}\right), \tag{3}$$

where $\mathbf{Y}_p^{(l)}$, $\mathbf{Z}_p^{(l)}$, and $b_p^{(l)}$ are the output map, input map, and bias of the 2D convolution of neuron p in layer l, respectively; $\mathbf{W}_{p,q}^{(l-1)}$ is the weight from neuron q in layer $l-1$ to neuron p in layer l; P_l is the number of neurons in layer l. The output of the activation function using sigmoid is given by

$$\mathbf{Z}_p^{(l)} = \sigma_{sig}(\mathbf{Y}_p^{(l)}) \tag{4}$$

with

$$\sigma_{sig}(x) = \frac{1}{1+e^{-x}}. \tag{5}$$

The average pooling layer follows the convolutional layer. The output takes the form of

$$\mathbf{Z}_p^{(l)}(r,c) = \frac{1}{4}\sum_{\alpha=0}^{1}\sum_{\beta=0}^{1}\mathbf{Z}_p^{(l-1)}(2r+\alpha, 2c+\beta), \tag{6}$$

where $\mathbf{Z}_p^{(l)}(r,c)$ denotes the (r,c)th element of the feature map. Because the STFT spectrogram has unequal length and width, the asymmetric kernel is used to tradeoff performance and complexity. Given the 17×9 kernel, the sizes of the output feature maps in the first convolutional layer and the first pooling layer are 112×12 and 56×6, respectively. The second convolutional layer and pooling layer then repeat the operations described by Eqs. 3, 4, as well as Eq. 6), and given the 5×5 kernel size, the feature maps of size 52×2 and 26×1 are generated. The outputs of the second pooling layer are flattened and the fully-connected layer

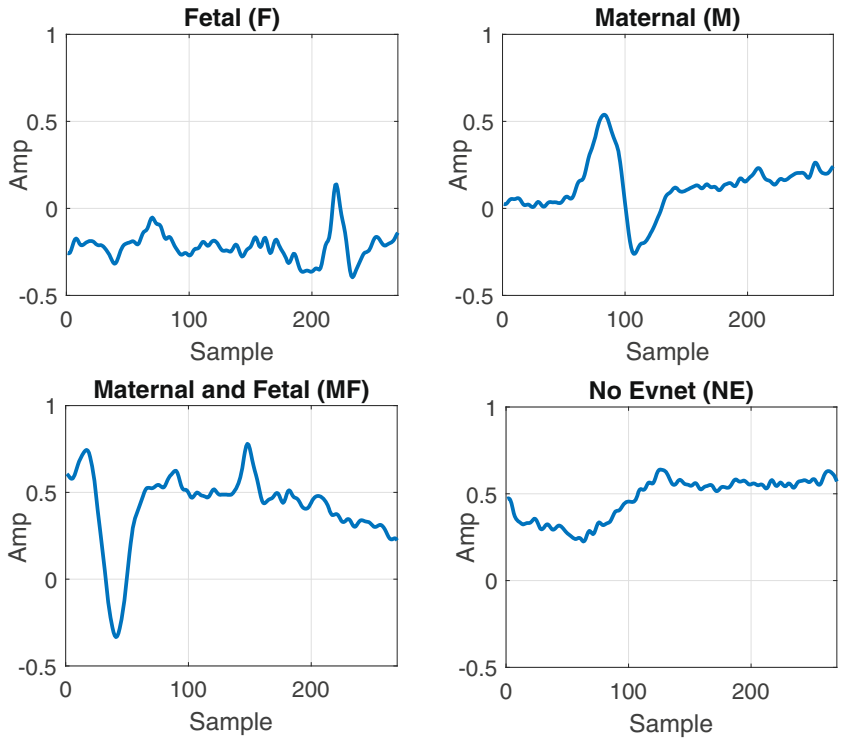


Figure 2 Waveform segments of the respective events.

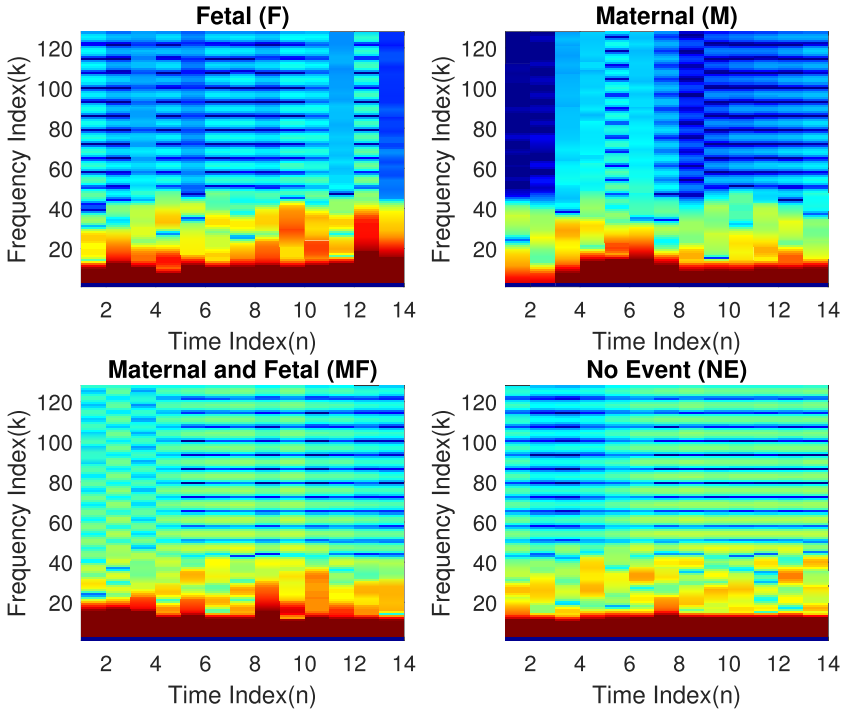


Figure 3 Time-frequency Spectrogram of the respective events.

denoted by $Z_q^{(L-1)}$, for $q = 0, 1, \cdots, 311$, is constructed. Finally, the output layer computes

$$Y_p^{(L)} = b_p^{(L)} + \sum_{q=0}^{P_{L-1}-1} W_{p,q}^{(L-1)} Z_q^{(L-1)}, \tag{7}$$

where $L(=6)$ denotes the index of the output layer and $P_{L-1} = 312$. The softmax activation function is adopted in the output layer. Let

$$\mathbf{Y}^{(L)} = [Y_0^{(L)} \quad Y_1^{(L)} \quad \cdots \quad Y_{P_L-1}^{(L)}],$$

and

$$\mathbf{Z}^{(L)} = \sigma_{sof}(\mathbf{Y}^{(L)}) = \frac{e^{\mathbf{Y}^{(L)}}}{\sum_{p=0}^{P_L-1} e^{Y_p^{(L)}}}, \tag{8}$$

where $\mathbf{Z}^{(L)} = [Z_0^{(L)} \quad Z_1^{(L)} \quad \cdots \quad Z_{P_L-1}^{(L)}]$ and $P_L = 4$. The probability of the pth event, $Z_p^{(L)}$, is obtained by squashing $Y_p^{(L)}$ into the range between 0 and 1.

In the training phase, denote the target as $\mathbf{T} = [T_0 \quad T_1 \quad \cdots \quad T_{P_L-1}]$, where $T_p \in \{0, 1\}$. The cross entropy is used as the loss function, which is given by

$$\xi = -\sum_{p=0}^{P_L-1} T_p \ln(Z_p^{(L)}). \tag{9}$$

The weight $W_{p,q}^{(l)}$ and bias $b_p^{(l)}$ of the 2D CNN are then adjusted by the stochastic gradient descent algorithm with the batch size equal to 8.

With the multi-channel abdominal ECG recordings, a fusion stage is designed to combine the classification results from different leads into one outcome. Given that the event

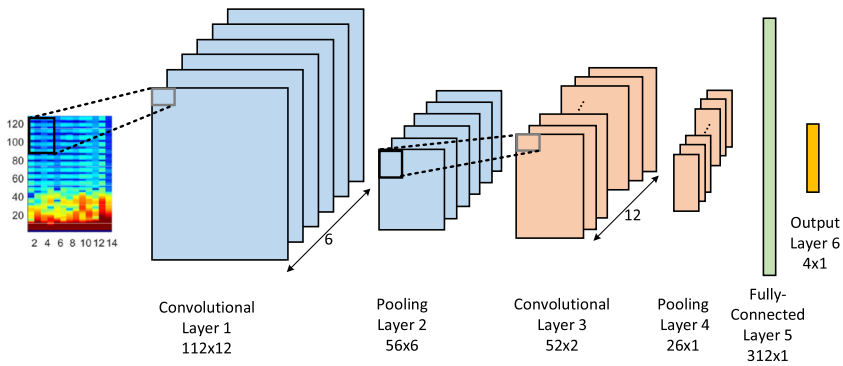


Figure 4 2D CNN structure for classification.

probability of lead i is represented by $\mathbf{Z}_i^{(L)} = [Z_{0,i}^{(L)} \quad Z_{1,i}^{(L)} \quad \cdots \quad Z_{P_L-1,i}^{(L)}]$, the fusion output is computed by

$$\mathbf{S} = [S_0 \quad S_1 \quad \cdots \quad S_{P_L-1}] = \frac{\sum_{i=1}^{4} f_i^3 \mathbf{Z}_i^{(L)}}{\sum_{i=1}^{4} f_i^3}, \tag{10}$$

where f_i is the fusion weight and is derived from the detection accuracy of the validation set. The final detection is given by

$$\mathcal{D} = \arg\max_p S_p. \tag{11}$$

3 Simulation Results

The publicly available PhysioNet database is then employed for performance verification [19]. The recordings from both the databases of the abdominal and direct fetal electrocardiogram and the PhysioNet/Computing in Cardiology Challenge 2013 are used.

For a one-minute recording, a total of 960 waveform segments are generated. The training data set and validation data set contain 392 and 180 segments, respectively. The testing data set has 388 segments. All are selected randomly. Because the width and length of the input spectrogram are not equal, Figure 5 first depicts the detection accuracies of subject r08 given different and asymmetric kernel sizes in the first convolutional layer. We can see that various detection accuracies in the training phase are obtained. Thus, a proper kernel width can examine the time span of a certain event in the spectrogram. The kernel size of 17×9 is used for its better performance.

The detection accuracies of the respective channels before fusion and after fusion are provided in Table 1. The weight proportional to f_i^3 in Eq. 10 derived from the accuracies of validation set is also given. We can see that the detection accuracy after fusion is significantly improved and the validation set plays an important role to fusion. The confusion matrix of subject a01 before fusion and after fusion is shown in Table 2. The fusion further improves the misjudgement of M event and MF event of channel 4.

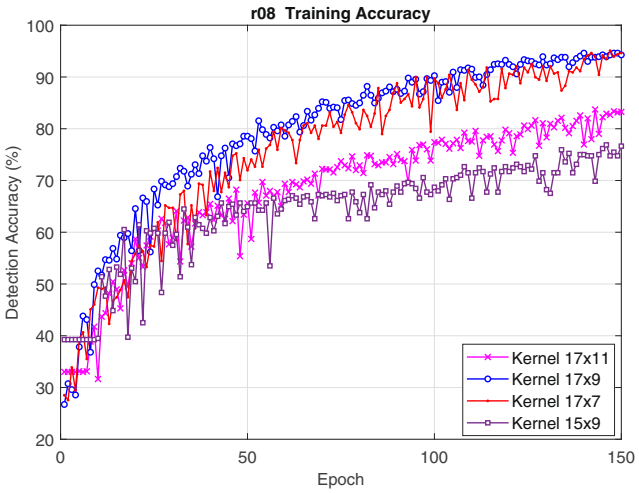


Figure 5 Accuracy comparison of the balanced and imbalanced kernel sizes.

The waveform segments from different pregnant women are then mixed to train the 2D CNN. We collect 2592 training segments, 720 validation segments, and 2448 testing segments from subjects r01, r08, and r10 of the abdominal and direct fetal electrocardiogram database. On the other hand, there are 1960 training segments, 900 validation segments, and 1940 testing segments from subjects a01-a05 of the PhysioNet/Computing in Cardiology Challenge 2013 database. The detection accuracies of testing set from two databases after fusion versus the number of epochs are depicted in Figure 6(a) and (b), respectively, which show 94.97% and 95.20% for the recordings from two databases.

The detection performance of different algorithms are then compared. In [16], Haar-wavelet transform (H-WT) and principle component analysis (PCA) are applied in the beginning and K-nearest neighborhood (KNN), support vector machine (SVM), and Bayesian network (BN) algorithms are employed for subsequent detection. The KNN algorithm achieves the best detection accuracy on the average. We adopt the H-WT, PCA, and KNN to detect four events here. In addition, we also apply the random forest algorithm that has 150 trees to the STFT spectrogram for detection. Table 3 shows the comparison of detection performance of mixed signal segments from subjects a01-a05 of the PhysioNet /Computing in Cardiology Challenge 2013 database. It is clear that the 2D-CNN with the STFT spectrogram input outperforms the other algorithms. Besides, the fusion enhances the detection accuracy apparently.

From Table 3, the advantage of applying 2D-CNN on the STFT spectrogram for fetal ECG detection is demonstrated. With detected fECG waveform segment, first, the famous Pan and Tompkin algorithm [20] can be employed for R peak detection and accordingly the time index of fetal R peak can be acquired to calculate R-R interval of the fetal ECG. Secondly, the waveform segment of fQRS (F event) and mQRS (M event) can be extracted for further ECG arrhythmia classification. Hence, the applicability of the event detection is broad.

4 Implementation of the Detector

We aim to implement the detector that is suitable for wearable devices. Figure 7 shows the hardware architecture of our design. The detector is composed of a spectrogram processor and a 2D CNN classifier. The spectrogram processor consists of a fast Fourier transform (FFT) module

Table 1 Detection accuracies of testing sets before and after fusion.

	r01		r08		r10		a02		a03		a04		a05	
	Acc.	Wei.	Acc.	Wei.	Acc.	Wei.	Acc.	Wei.	Acc.	Wei.	Acc.	Wei.	Acc.	Wei.
Ch. 1	74.8%	0.260	74.9%	0.283	78.4%	0.307	86.2%	0.294	80.6%	0.280	86.5%	0.282	96.8%	0.348
Ch. 2	65.1%	0.231	60.0%	0.203	75.4%	0.307	78.7%	0.110	95.3%	0.363	91.4%	0.333	71.5%	0.201
Ch. 3	79.3%	0.171	79.7%	0.212	48.5%	0.058	79.0%	0.230	68.2%	0.100	60.3%	0.167	76.8%	0.257
Ch. 4	94.2%	0.339	88.6%	0.302	73.7%	0.329	95.9%	0.365	79.9%	0.258	69.9%	0.219	59.3%	0.193
Fusion	94.8%		96.8%		86.0%		98.3%		96.0%		94.1%		98.7%	

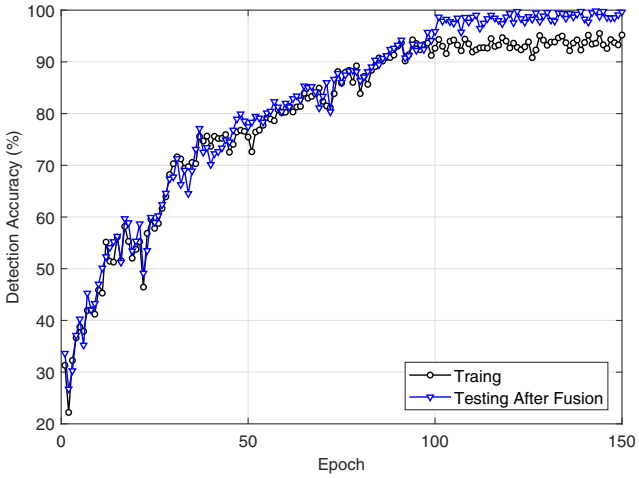
and a module for absolute function. The 2D CNN classifier computes 2D convolution, activation function, and average pooling. Details will be presented in the following.

4.1 Spectrogram Processor

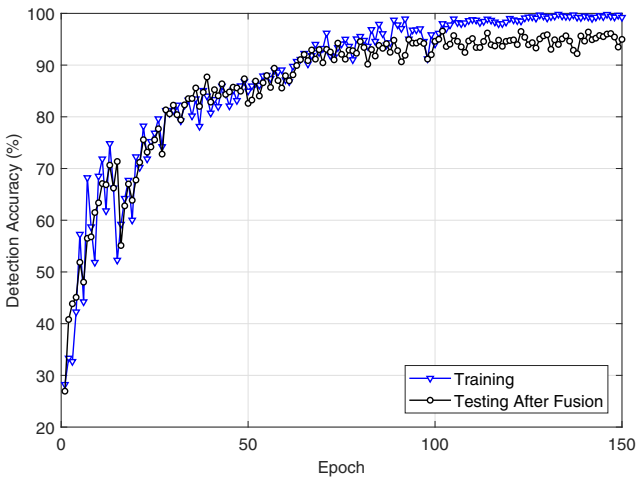
The ECG recordings are saved into input Memory sequentially. Once that four-channel signals of one segment are ready, the FFT module is activated. To accomplish the STFT spectrogram, multiple FFT operations are required. Because the ECG recording is a real signal, the packing and doubling algorithms are considered [21]. For M real

Table 2 Confusion matrixes of testing set from the respective channels and the result after fusion.

Ground Truth	F Event	M Event	MF Event	N Event
Classification Results of Abdomen 1				
F Event	36	4	8	1
M Event	5	20	9	1
MF Event	8	8	25	0
N Event	13	3	0	63
Classification Results of Abdomen 2				
F Event	50	12	12	10
M Event	9	20	11	3
MF Event	2	1	16	2
N Event	1	2	3	50
Classification Results of Abdomen 3				
F Event	50	6	1	5
M Event	3	25	3	1
MF Event	8	2	38	0
N Event	1	2	0	49
Classification Results of Abdomen 4				
F Event	61	1	1	0
M Event	1	32	6	0
MF Event	0	1	33	0
N Event	0	1	2	65
Classification Results After Fusion				
F Event	61	1	2	0
M Event	1	33	2	0
MF Event	0	1	38	0
N Event	0	0	0	65



(a)



(b)

Figure 6 Detection accuracy of mixed waveform segments from (a) the abdominal and direct fetal electrocardiogram database, and (b) the PhysioNet/Computing in Cardiology Challenge 2013 database.

Table 3 Comparison of different detection algorithms.

Algorithm	Spectrogram + 2D-CNN (Fusion)	Spectrogram + 2D-CNN	Spectrogram + Random Forest	H-WT + PCA + KNN [16]
Accuracy (%)	95.20	88.48	86.22	69.65

samples, the packing algorithm combines the even-indexed and odd-indexed samples into $M/2$ complex data and performs $M/2$-point FFT. Extra additions and multiplications are demanded to restore the spectrum in the positive frequency band. On the other hand, the doubling algorithm forms M complex data by two sets of M real samples and uses M-point FFT. Only additions are needed for further separating their respective spectra. Considering the arithmetic complexity and the multi-channel recordings, we adopt the doubling algorithm.

In the doubling algorithm, the recordings of $2i$-th and $(2i+1)$-th channels are gathered to form the complex data, which is given by

$$d(n,m) = (x_{2i}(m+n\Delta) + jx_{2i+1}(m+n\Delta))\, w(m), \quad m = 0, 1, \cdots, M-1. \tag{12}$$

After M-point FFT, we have

$$D(n,k) = \mathbf{X}_{2i}(n,k) + j\mathbf{X}_{2i+1}(n,k), \quad k = 0, 1, \cdots, M-1. \tag{13}$$

Because the spectrum of real time-domain signals is conjugate symmetric, $X_{2i}(n,k)$ and $X_{2i+1}(n,k)$ then take the form of

$$\begin{aligned} \mathbf{X}_{2i}(n,k) &= \tfrac{1}{2}\left(D(n,k) + D^*(n,M-k)\right) \\ \mathbf{X}_{2i+1}(n,k) &= -\tfrac{j}{2}\left(D(n,k) - D^*(n,M-k)\right). \end{aligned} \tag{14}$$

The signals $\mathbf{X}_{2i}(n,k)$ and $\mathbf{X}_{2i+1}(n,k)$ in the positive frequency band, namely $k = 0, 1, \cdots, M/2$, can be acquired by M complex additions.

As shown in Figure 7, real signals of channels $2i$ and $2i+1$ are fetched from the input memory and are sent into the FFT module, which is realized by the radix-2^2 single-path delay feedback architecture [22]. The FFT output $D(n,k)$ is written back into the memory using the bit-reversed address. In order to derive the respective spectrogram of $\mathbf{X}_{2i}(n,k)$ and $\mathbf{X}_{2i+1}(n,k)$ in the positive frequency band, $D(n,k)$ and $D(n,M-k)$ are read from the memory successively and the separation module computes (14). Then, $|\mathbf{X}_{2i}(n,k)|$ and $|\mathbf{X}_{2i+1}(n,k)|$ are calculated by the magnitude module, which is implemented by the coordinate rotation digital computer (CORDIC), and are saved into the input map memory for subsequent processing by the 2D-CNN classifier.

4.2 2D-CNN Processor

Various CNN processors have been implemented in the literature. The hardware utilization and power efficiency are important metrics of a CNN processor [23]. Usually, the processor possesses processing element (PE) arrays to accelerate computations by parallel processing. Higher hardware utilization relies on the mapping of the computations to the convolution core. High power efficiency can be obtained by well scheduling of data or weight reuse to reduce the frequent memory accesses. The famous deep neural network processors all show their superiority of reconfigurability to accommodate various models with good hardware efficiency and data or weight reuse [24–26]. In [23], the memory access times of different reuse strategies including input reuse, output reuse, and weight reuse, are analyzed. The input feature map is fully utilized

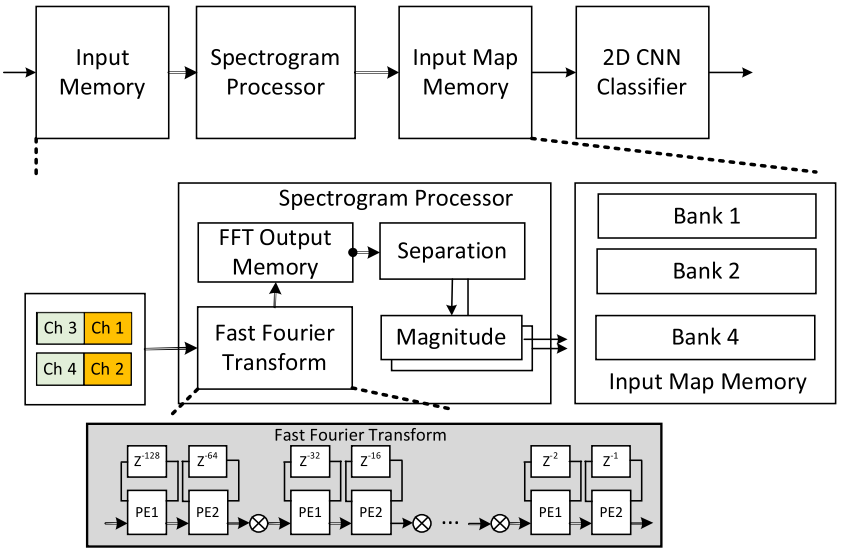


Figure 7 Architecture of the detector.

for calculating partial sums in the input reuse strategy. In the output reuse strategies, the temporal accumulation of the partial sums is saved in the output buffer until the output map is completed. The kernel weights in the weight buffer are exploited completely to realize weight reuse strategy. Input-output mapping with spatial accumulation and spatial mapping with temporal accumulation are illustrated in [26].

However, unlike the prior works, the 2D CNN classifier in this detector is dedicated to the ECG spectrogram classification and has the following distinct features.

- The input and output feature maps associated with the spectrogram has a narrower width in the time domain compared to the height in the frequency domain.
- The kernel size is large and also asymmetric to capture features spanning different ranges in the frequency and time domains.

A hybrid strategy taking advantage of input reuse and output reuse as well as spatial mapping with temporal accumulation is then adopted for different convolution layers to enhance the PE utilization and to reduce the hardware complexity.

Figure 8 depicts the architecture of the 2D-CNN classifier. Six PEs are incorporated, each having two sets of 1D convolution core. One adder is inserted to combine the partial sums and the combined results are saved in the feature map memory. The data of the map memory are processed by the sigmoid function and pooling function and then sent back to the map memory. The second convolution that is accomplished by the same computation blocks repeats. The fully-connected layer is realized by the multiplier-and-accumulator (MAC). The softmax function then computes the respective probabilities of four events. The major layers will be described in the following.

4.2.1 First Convolution Layer

For the first convolution layer, the input map of size 128×20 with padding zeros is convolved with six kernels of size 17×9 to generate six output maps of size 112×12. Each PE handles one output feature map. The input reuse strategy is adopted and the kernel is decomposed into $17\times$ 1D convolution. Because each PE contains two sets of 9-tap 1D convolution core, two adjacent rows of the input feature map are fetched from the input map memory. The odd-indexed and even-indexed rows are sent to the top and bottom convolution cores, respectively.

Usually the latency of the convolution core is determined by the processing kernel size. If the feature map size is much larger than the kernel size, just like the common cases of the prior works, the impact of the latency is almost negligible. However, if the feature map size is close to the kernel size, such as this case, the latency caused by data buffering will result in low PE utilization significantly. Because of the long-latency of the 9-tap convolution core, a delay buffer with a pre-fetch mechanism is designed to maintain the high PE utilization. As shown in Figure 8, the delay buffer consists of shift registers A and B (SRA and SRB). While SRA supplies the data for convolution, SRB prepares the data for convolution involving with the next row of the 2D input map or with the next row of the 2D kernel weight and vice versa.

Figure 9 shows the example of the data flow in the SRA and SRB. At cycle A, the convolution core performs the last convolution of the kernel row 1 and the input map row 1 which is sent from the input map memory. Meanwhile, the fetch of input map row 3 has already started and some data have existed in the SRB. At the next cycle B, the pre-fetched data from input map row 3 are ready and the core switches to use the data in the SRB for the convolution with row 3 of the kernel. At some cycle C, the convolution of the input map row 3 and kernel row 3 is completed. At the next cycle D, the core switches to use the data in the SRA, where the data of input map row 3 are preserved with the circulation in the SRA and SRB, to compute the convolution with kernel row 2. Similarly, from cycle E to cycle F, the core performs convolution of kernel row 1 and input map row 3 by utilizing the data in SRA. With the proposed pre-fetch mechanism, seamless convolutions in the PEs can be achieved.

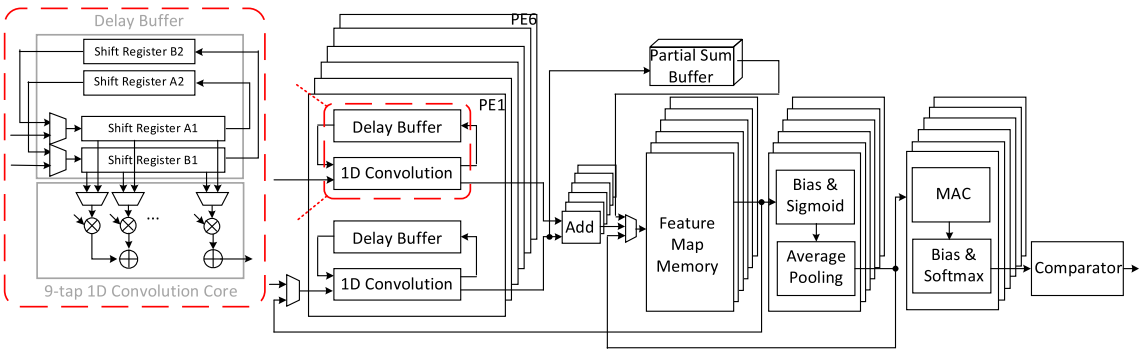


Figure 8 Architecture of the 2D-CNN classifier.

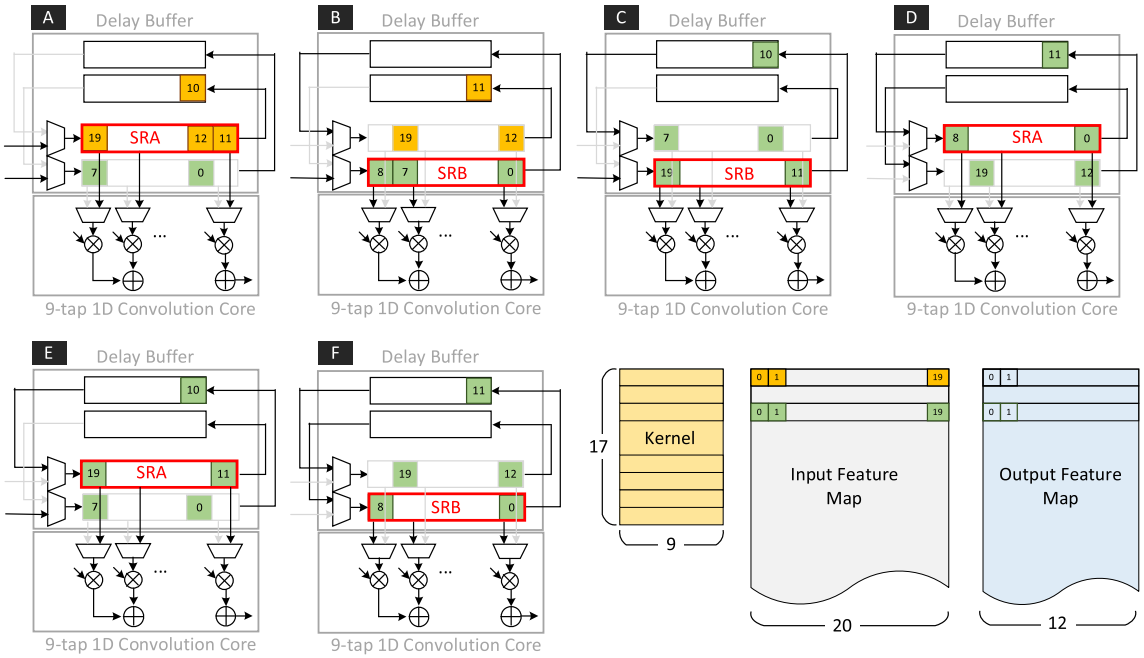


Figure 9 Shift register flow with a pre-fetch mechanism.

Given the kernel size $K_y \times K_x$ and the input feature map size $I_y \times I_x$, the total number of weight updates for 2D convolution realized by 1D convolution core ($1 \times K_x$) can be expressed as

$$\begin{aligned}(I_y - 2(K_y - 1))K_y + 2(1 + 2 + \cdots + K_y - 1) \\ = K_y(I_y - K_y + 1) = K_y \cdot O_y,\end{aligned} \tag{15}$$

where $O_y \times O_x$ is the size of output feature map. Although two 1D convolution cores are implemented, the kernel height is an odd number. With the proposed pre-fetch mechanism, the number of total computation cycles is calculated by $O_y \cdot O_x \cdot (K_y + 1)/2$. The hardware utilization can be evaluated by

$$\frac{O_y \cdot O_x \cdot K_y}{O_y \cdot O_x \cdot (K_y + 1)} = \frac{K_y}{K_y + 1}. \tag{16}$$

In our case, the utilization achieves 94.4%. If no pre-fetch mechanism is adopted, it will take $O_y \cdot I_x \cdot (K_y + 1)$ clock cycles and the utilization will becomes

$$\frac{O_y \cdot O_x \cdot K_y}{O_y \cdot I_x \cdot (K_y + 1)} = \frac{O_x K_y}{I_x (K_y + 1)}, \tag{17}$$

which is 56.7%. Hence, our proposed pre-fetch mechanism improves the hardware utilization for the computations of the first convolution layer significantly.

4.2.2 Second Convolution Layer

After adding the bias to the output feature map, the processor calculates the sigmoid function output of the respective data. The sigmoid function is realized by table lookup and its symmetry is utilized to save the table size. The average pooling is accomplished by the shifter. The feature map size in the memory becomes 56×6. Processing of the second convolution then starts with the kernel of size 5×5. The same PEs and feature map memory are utilized for the computation.

Because of hardware sharing, the feature map memory not only needs to provide the input of the second convolution but also needs to be the the storage for accumulating the intermediate convolution results. Due to the seamless computation with switching between the SRA and SRB as mentioned in Sec. 4.2.1), the number of memory ports is not sufficient. To avoid allocating more memory modules, partial sum buffers are designed for output reuse. The 1D convolution outputs from 6 PEs are summed up first and then the partial sums are saved into different buffers according to their kernel index and row index as shown in Figure 10. In the second convolution, the kernel size is 5×5. The processor has to accumulate five 1D convolution results and the sequence of computations that follows the input reuse strategy is also indicated in Figure 10. Hence, only $K_y \times O_x (= 5 \times 2)$ local buffers are required for the accumulation. Once the complete 2D convolution result is accumulated, it is written into the feature map memory. Consequently, two-port memory can be used and shared for saving the feature maps of layer 1 to layer 4. Both input reuse and output reuse are achieved for the second convolution layer with only small partial sum buffers.

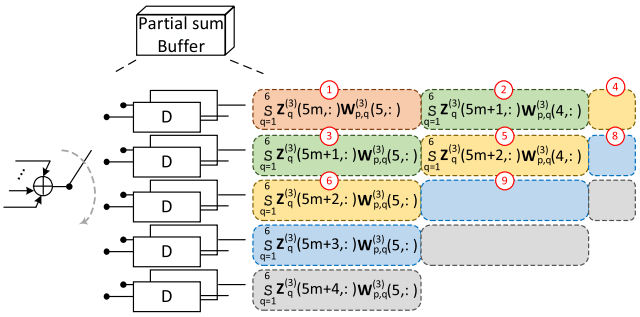


Figure 10 Block diagram of the partial sum buffer and its operation.

4.2.3 Fully-Connected Layer and Softmax Function

The computation for the fully-connected layer, which generates four outputs corresponding to the four events, is simpler. Six multiplier-and-accumulators (MACs) are implemented. Note that the output feature map of layer 4 is small compared to the input feature map of layer 1. If we properly allocate the feature map in the memory, the next processing for 2D convolution of layer 1 can start by utilizing the released PEs.

The softmax function is required to generate the associated probability of each event for fusion. The alternative form mentioned in [27] is used,

$$\mathbf{Z}^{(L)} = \sigma_{sof}(\mathbf{Y}^{(L)}) = \frac{e^{\mathbf{Y}^{(L)} - max(\mathbf{Y}^{L})}}{\sum_{p=0}^{P_L-1} e^{Y_p^{(L)} - max(\mathbf{Y}^{(L)})}}, \tag{18}$$

and thus the table size can be reduced with the input offset. Finally, a comparator is used to determine the event with the highest probability.

5 Hardware Implementation Results

The proposed detector is implemented on the FPGA platform. The fixed-point representation is adopted for the datapath. The input of the ECG recording and the twiddle factors are quantized to 11 bits and 10 bits, respectively. The STFT output is represented by 12 bits. We use 17 bits for the outputs of the convolution layer and 16 bits for the outputs of the fully connected layer. Figure 11 shows the simulation results of the fixed-point representation. The degradation is almost negligible. In Table 4, the utilization of FPGA resources is provided. Five block RAMs are used. The distributed RAMs are adopted for feature map memory. There are 126 DSP48E1s. Among them, 108 multipliers are utilized in PEs for convolution, 6 multipliers are used for MACs, and the remaining ones exist in the spectrogram processor. According to the synthesis report, the highest operating frequency can achieve 70.8MHz.

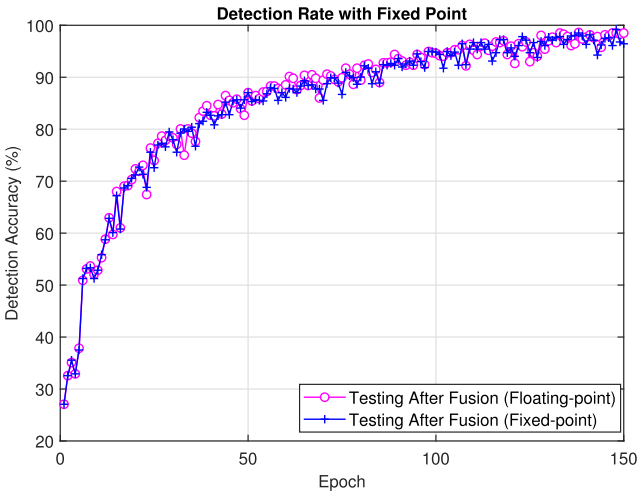


Figure 11 Simulation result of fixed-point representation.

The FFT needs to perform 14 times in order to generate the spectrogram. Because of the doubling algorithm and the pipeline architecture, the frequency domain data of two channels can be generated every 528 clocks. A total of 8160 clock cycles are required to finish the computations of spectrograms of two channels. Then, as mentioned in Eq. 16, it takes 12105 clock cycles to complete the 2D convolution in the first convolution layer, which is the most time-consuming operation. On the other hand, the second convolution layer needs 6388 clock cycles for the computation. If the proposed pre-fetch mechanism is not applied, due to the latency caused by buffering the data of the new row, the computation times for the first and the second convolution layers become 20169 cycles and 18868 cycles. On the other hand, with the allocation of 10 partial sum buffers indicated in Figure 10, we can eliminate additional feature map memory of size $52 \times 2 \times 12$. The design improvements are summarized in Table 5.

The classifier needs 20258 clock cycles to process one spectrogram. Since the spectrogram processor and the 2D CNN processor can be activated simultaneously, a total of 81032 clock cycles are required in order to derive a fusion result for one waveform segment. Thus, for real-time

Table 4 List of FPGA utilization.

Device	7vx690t	
Number of Slice Registers	8661	<1%
Number of Slice LUTs	31455	7%
Number used as logic	26091	6%
Number used as memory	5364	3%
Number of Block RAM/FIFO	5	<1%
Number of DSP48E1s	126	3%

Table 5 Design improvements.

	Without Improvement	With Improvement
Pre-Fetch Mechanism	Convolution Layer 1 20169 (cycles) (100%)	12105 (cycles) (49.6%)
	Convolution Layer 3 18868 (cycles)	6388 (cycles)
	Classifier Total 40802 (cycles)	20258 (cycles)
Partial Sum Buffer	Convolution Layer 3 1248 (words)	10 (words)
	Total Data Memory 38.6KB (100 %)	35.9KB (93.0 %)

processing, the clock frequency can be lowered to 324KHz to save power consumption. For offline processing, the detector can generate 864 fusion outputs every second. An ASIC realizing BLSTM and 1D CNN in 40nm CMOS technology is described in [28]. It takes 130533 clock cycles to derive inference result with 100MHz clock frequency. In Table 6, we list the implementations with CNN on FPGA platform. However, it is not easy to make a fair comparison. The complexity and the target applications are quite different. For object classification, higher throughput with power efficiency is often desired. Besides CNN, an FFT processor is also included in our design to acquire the spectrogram. Our work shows a lightweight design with small memory requirements, reasonable inference clock cycles, and low power consumption for real-time abdominal fetal ECG Detection.

6 Conclusion

To detect fetal ECG signals from multi-lead abdominal ECG recording, we propose to use the 2D CNN approach on the spectrogram. Hence, in the preprocessing phase, the recording is normalized and partitioned into segments. Then, STFT is applied with proper windowing. Two 2D CNN layers and one fully-connected layer are used for the classification. Finally, fusion is performed that applies weighting derived from the detection accuracy of the validation set to combine the classification outputs of four channels. From simulation result, the proposed scheme achieves detection accuracy of 95.2% and outperforms the conventional one. To verify its feasibility in the wearable device, the detector is implemented on the FPGA platform. It consists of the spectrogram processor and the 2D CNN classifier. Because the two dimensions of the spectrogram and the kernel size are asymmetric, we propose a pre-fetch mechanism to support seamless computations during 2D convolution that is mapped onto 1D convolution core. The long latency caused by buffering new data for convolution can be eliminated. In addition, a partial sum buffer is allocated so that the feature map memory can be shared and the flow exploits input reuse and output reuse. It takes 20258 clock cycles for one inference and uses 41.3KB memory. The effectiveness of the pre-fetch mechanism and allocation of the partial sum buffer is demonstrated. The design also shows the feasibility of a lightweight 2D CNN approach for fetal ECG detection.

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Table 6 List of ECG related classifiers.

	This work	[29]	
Input Wordlength (bits)	12	16	8
Output Wordlength (bits)	16	16	8
Function	Spectrogram + 2D CNN	2D CNN	
Applications	Fetal ECG Detection	Object Detection	
Implementation	FPGA (Virtex7)	FPGA (Zynq)	
Inference (cycles)	20258	-	
Memory (KByte)	41.3	-	
Clock (MHz)	70.8	150	214
DSP (for CNN)	108	780	190
Power (W)	15.33×10^{-3} (@324KHz,1V)	9.63(@150MHz)	3.5(@214MHz)

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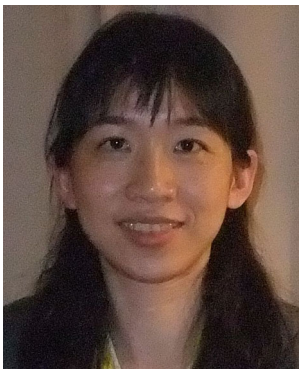
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